



## Homework 5

**Deadline: 2 November 2025, 23:30.**

*All solutions must be in a single PDF file and uploaded to the LMS portal.*

1. Consider the following estimator of volatility:

$$\sigma_t^\pi = \sqrt{\frac{1}{\tau} \sum_{k=1}^n (\Delta \ln S_{t_k})^2}$$

where  $\pi = (t_0, \dots, t_n)$  is a partition of the time segment  $[t - \tau, t]$ , and the price of the underlying asset follows the process

$$d \ln S_t = b_t dW_t.$$

- (1 point) Derive expressions for  $\text{Bias}(\sigma_t^\pi)$  and  $\text{Var}(\sigma_t^\pi)$  for the case of piecewise constant volatility, i.e.,  $b_t$  is constant on each  $[t_{k-1}, t_k]$ .
- (0.5 point) Analyze how  $\text{Bias}(\sigma_t^\pi)$  and  $\text{Var}(\sigma_t^\pi)$  change under the following conditions:
  - The partition size  $\|\pi\|$  decreases while  $\tau$  is held fixed;
  - Time length  $\tau$  increases while  $\|\pi\|$  is held fixed.

2. Suppose that at the initial time we have estimated  $\sigma(t)$  for the model

$$dS_t = rS_t dt + \sigma(t)S_t dW_t.$$

Now consider two option pricing models:

- The modified Black-Scholes-Merton model<sup>1</sup> with the underlying asset's process using the estimated  $\sigma(t)$ ;
- The Black-Scholes-Merton model with constant volatility  $\sigma = \sqrt{\frac{1}{T} \int_0^T \sigma^2(u) du}$ .

Your tasks are to determine and explain:

- (0.5 point) Will the option's delta computed at the initial time differ between these models?
- (0.5 point) Will it differ at an arbitrary time  $t > 0$ ?
- (0.5 point) Will the option's theta differ between these models?

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<sup>1</sup>See Lecture 5, Section 2.